Saju Khakurel *Embedded Systems, Computing & Firmware*

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Blogs | Website

Profile

Final-year Erasmus Mundus MSc student with experience in firmware development, system integration, and industrial IoT architectures. Strong background in control systems, signal processing, and real-hardware validation. Currently working on industrial IIoT platform evaluation and open-source stack prototyping.

Education

Erasmus Mundus Joint Master of Embedded Intelligent Nanosystems Engineering,
University of Orléans (France) • University of Siegen (Germany) • Hellenic Mediterranean University (Greece)
2024/10 – Present

Bachelor's Degree in Electronics & Communication Engineering,
Tribhuvan University, Pulchowk Campus
2017 – 2022 | Nepal

Master's Thesis

Comparative Analysis of IIoT Platforms & Open-Source Stack Prototype, University of Siegen | In collaboration with EJOT SE & Co. KG
2026/03 – Present | Siegen, Germany
- Evaluating industrial IoT platforms (ThingWorx reference) across security, scalability & interoperability
- Implementing a modular open-source IIoT stack for a comparative PoC

Skills

Programming (C/C++, Python, MicroPython, SQL)

Embedded Systems (STM32, ESP32, Arduino, Raspberry Pi, FreeRTOS, OTA, BLE, CAN, UART, SPI, I²C, debugging)

Robotics & Control (ROS, SLAM, MATLAB, Simulink, automation, control systems, Kalman filtering)

IoT & Data (MQTT, LoRa, InfluxDB, Grafana, databases)

Signal Processing & AI (DSP, estimation theory, Fourier analysis, deep learning)

Tools & Environments (Linux, Git, Octave, LTspice, VS Code)

Hardware & Electronics (KiCad, PCB design, soldering, sensors, actuators, motors, op-amps, MOSFETs, oscilloscopes, encoders)

Professional Experience

Embedded Systems Engineer — Vehicle Software, Yatri Design Studio
2021/12 – 2024/09 | Nepal
- Developed embedded C/C++ firmware for STM32-based systems used in electric vehicles.
- Worked on firmware features involving device startup, communication interfaces, and OBD diagnostics.
- Supported the development and validation of OTA software update workflows for embedded devices and charging stations.
- Performed firmware testing, debugging, and log analysis on real hardware before deployment.
- Collaborated with firmware, hardware, and backend teams during integration and testing phases.
- Development of the Internal Server + OTA in 60kW charging stations

Hardware Designer and Embedded System Programmer,
Robotics Club, Pulchowk Campus
2018 – 2020 | Lalitpur, Nepal
- Designed and tested PCBs for mobile robots used in national robotics competitions.
- Developed embedded firmware for sensor integration and encoder-based motion correction

Avionics Systems Designer, Team Member, NEAR Aerospace
2020 | Nepal
- Designed and implemented a microcontroller-based data-logging system

Academic Projects

Connected Greenhouse IOT
Built multi-task IoT firmware with queue-based communication | I²C sensors + OLED UI + Rotary encoder | MQTT⇌Node-RED⇌Grafana | remote monitoring & control.

TinyML-Based Real-Time Activity Recognition on Memory-Constrained Smartwatch
Designed and deployed deep learning models for real-time activity recognition on the Bangle.js 2 (256KB RAM) and implemented TensorFlow Lite quantization for edge deployment

Automatic Modulation Classifier using DL
Built DL-based LSTM models to classify modulation types in noisy signals; explored Bi-LSTM and attention layers for improved accuracy.

NAST-Funded Disinfection Robot ,
Protoype for quarantine areas during COVID-19
Designed embedded hardware and firmware for a Bluetooth-controlled spraying system, focusing on reliable actuator control.

SLAM (TinySLAM) under ROS
Implemented particle filter SLAM with simulation & visualization in ROS (rviz/gazebo).

Rockbye Baby
Developed a sensor module using a 555-timer-based humidity detection. | Awarded LOCUS 2020 Winner.

Publications

Book Contribution
- Maharjan, Asim, and Saju Khakurel. "Introduction to IoT." IoT, Machine Learning and Blockchain Technologies for Renewable Energy and Modern Hybrid Power Systems, edited by C. Sharmeela et al., River Publishers, 2022, pp. 1–23

Languages

English C1 | Nepali | Hindi | French A1 | German A1.1

Volunteering & Leadership

Instructor & Technical Mentor (Voluntary), 2019 – 2024
Robotics Club Pulchowk Campus • Locus • Godawari Residential School • Skill Tour Nepal
- Taught Arduino, sensors, motors, PCB design, and embedded programming to university and school students.
- Mentored student teams for robotics competitions and science exhibitions.
- Introduced basic programming and robotics concepts to school children through hands-on workshops.

EV Awareness & Outreach – Yatri Energy (USAID-supported) 2024/04 – 2024/05
- Led EV awareness sessions and public discussions across major Nepali cities, delivering technical presentations and interactive demonstrations. Nepal